

MASTER TEST PLAN AND REPORT

Kelompok 7 - EduScale

31 Juli 2026

Contents

MASTER TEST PLAN AND REPORT	1
EduScale – School Management System	1
BAGIAN A – MASTER TEST PLAN	2
A.1 Pendahuluan & Tujuan	2
A.2 Ruang Lingkup Pengujian (Scope)	2
A.3 Strategi Pengujian	2
A.4 Kriteria Masuk & Keluar Pengujian	2
A.5 Sumber Daya Pengujian	3
A.6 Jadwal Pengujian	3
A.7 Tools dan Environment	3
A.8 Risk Assessment	4
BAGIAN B – TEST REPORT (HASIL PENGUJIAN)	4
B.1 Ringkasan Eksekutif	4
B.2 Hasil per Level Pengujian	4
B.3 Hasil Unit Testing	5
B.4 Hasil Integration Testing	5
B.5 Hasil System Testing (E2E)	6
B.6 Hasil Black-Box Testing (EP & BVA)	6
B.7 Code Coverage	7
B.8 Load Testing – Konfigurasi & Target	7
B.9 API Automation (Postman/Newman)	8
B.10 UI Automation (Playwright)	8
B.11 Defect Summary	8
B.12 Kesimpulan	8

MASTER TEST PLAN AND REPORT

EduScale – School Management System

Nama Proyek: EduScale School Management System

Nomor Dokumen: MTP-EDUSCALE-001

Versi: 1.0

Tanggal: 31 Juli 2026

Kelompok: 7

Mata Kuliah: Advanced Software Testing and Quality Assurance (ASTQA)

Status: FINAL

BAGIAN A – MASTER TEST PLAN

A.1 Pendahuluan & Tujuan

Dokumen ini merupakan Master Test Plan sekaligus Test Report untuk pengujian sistem EduScale. Tujuannya: 1. Mendefinisikan strategi dan ruang lingkup pengujian 2. Menetapkan sumber daya, jadwal, dan kriteria kelulusan 3. Melaporkan hasil eksekusi seluruh level pengujian

A.2 Ruang Lingkup Pengujian (Scope)

In Scope: - Seluruh modul backend API (Node.js/Express) - Seluruh halaman frontend (React SPA) - Alur autentikasi dan otorisasi - Operasi CRUD seluruh entitas (Siswa, Guru, Kelas, Mapel, Jurnal, BK) - Dashboard dan statistik - Performa dan beban sistem

Out of Scope: - Infrastruktur deployment (server production) - Pengujian browser mobile native - Pengujian database backup/recovery

A.3 Strategi Pengujian

Level	Pendekatan	Tool	Teknik
Unit Testing	White-box, isolasi fungsi	Jest v30	Assertion, Mocking
Integration Testing	API testing, in-memory mock	Jest + Supertest	Black-box API
System Testing	E2E flow lengkap	Jest + Supertest	Skenario bisnis
Acceptance Testing	UAT manual	Browser	Kualitatif
Black-Box Testing	EP & BVA	Jest + Supertest	Equivalence Partitioning, BVA
Load Testing	Stress & spike	k6	Concurrent users
API Automation	Collection runner	Postman/Newman	Scripted assertions
UI E2E Automation	Browser automation	Playwright	Page Object, Scenario

A.4 Kriteria Masuk & Keluar Pengujian

Entry Criteria (Pengujian Dapat Dimulai): - [] Source code telah di-commit ke repository - [] Backend API dapat dijalankan (npm start) - [] Frontend dapat dijalankan (npm run dev) - [] Database seed berhasil dijalankan - [] Dependensi testing telah terinstall

Exit Criteria (Pengujian Dinyatakan Selesai): - [] Semua test case otomatis (Jest) PASS $\geq 95\%$
 - [] Code coverage $\geq 70\%$ pada modul kritis - [] Semua test case CRITICAL di UAT PASS - []
 Load test memenuhi threshold performa - [] Tidak ada defect Severity 1 (Critical) yang terbuka

A.5 Sumber Daya Pengujian

Peran	Tanggung Jawab
Test Lead	Koordinasi, review test plan, laporan akhir
Backend Tester	Unit test, integration test, API testing
Frontend Tester	Playwright E2E, UAT
Performance Tester	Load test, analisis hasil

A.6 Jadwal Pengujian

Aktivitas	Target Selesai
Penyusunan SRS & SDD	Minggu 1
Unit & Integration Testing	Minggu 2
Black-Box Testing (EP & BVA)	Minggu 2
System Testing (E2E)	Minggu 3
Acceptance Testing (UAT)	Minggu 3
Load & Automated Testing	Minggu 3-4
Penyusunan Laporan	Minggu 4

A.7 Tools dan Environment

Komponen	Tool/Versi
Unit/Integration	Jest v30.4.2 + Supertest v7.2.2
Black-Box	Jest v30.4.2 + Supertest v7.2.2
UI E2E	Playwright v1.45+
API Automation	Postman v11 / Newman CLI
Load Testing	k6 v0.50+
OS	macOS
Node.js	v18+
Database	PostgreSQL ≥ 13

A.8 Risk Assessment

Risiko	Probabilitas	Dampak	Mitigasi
Database tidak tersedia saat test	Sedang	Tinggi	Gunakan in-memory mock untuk unit/integration test
Playwright tidak bisa menemukan elemen UI	Tinggi	Sedang	Gunakan multiple selectors, tambah retry
Load test membebani server lokal	Rendah	Sedang	Batasi VU, gunakan smoke test dulu
Perubahan kode saat pengujian	Sedang	Tinggi	Freeze kode saat eksekusi test

BAGIAN B – TEST REPORT (HASIL PENGUJIAN)

B.1 Ringkasan Eksekutif

Metrik	Nilai
Total Test Cases Otomatis	172
PASSED	172
FAILED	0
Tingkat Keberhasilan	100%
Total Test Suites	6
Waktu Eksekusi	~2.4 detik
Tanggal Eksekusi	31 Juli 2026

B.2 Hasil per Level Pengujian

HASIL PENGUJIAN EDUSCALE – SEMUA LEVEL				
Level	Total	PASS	FAIL	%
Unit Testing	55	55	0	100%
Integration Testing	30	30	0	100%
System Testing (E2E)	22	22	0	100%
Black-Box (EP & BVA)	65	65	0	100%
Acceptance Testing (UAT)	19	Manual	Manual	Ready
Load Testing (k6)	–	–	–	Ready

API Automation (Newman)	28	–	–	Ready
UI E2E (Playwright)	26	–	–	Ready
TOTAL OTOMATIS (Jest)	172	172	0	100%

B.3 Hasil Unit Testing

File: eduscale-backend/tests/unit/

Framework: Jest v30

Hasil: 55/55 PASSED (100%)

Suite	File	TC	Hasil
generateToken	utils.test.js	6	✓ PASS
Paginasi Logic	utils.test.js	5	✓ PASS
Password Validation	utils.test.js	4	✓ PASS
Input Validation	utils.test.js	4	✓ PASS
authMiddleware	middleware.test.js	5	✓ PASS
roleMiddleware	middleware.test.js	5	✓ PASS
HTTP Status Codes	middleware.test.js	6	✓ PASS
Auth Controller	controllers.test.js	4	✓ PASS
Student Controller	controllers.test.js	6	✓ PASS
Teacher Controller	controllers.test.js	3	✓ PASS
Dashboard Controller	controllers.test.js	3	✓ PASS
BK Controller	controllers.test.js	4	✓ PASS
TOTAL		55	✓ 100%

B.4 Hasil Integration Testing

File: eduscale-backend/tests/integration/api.test.js

Framework: Jest + Supertest

Hasil: 30/30 PASSED (100%)

Suite	TC	Hasil
IT-01: Base API	1	✓
IT-02: Authentication	7	✓
IT-03: Students CRUD	10	✓
IT-04: Teachers	4	✓
IT-05: Dashboard	4	✓
IT-06: JWT Security	3	✓
IT-07: Pagination	1	✓
TOTAL	30	✓ 100%

B.5 Hasil System Testing (E2E)

File: eduscale-backend/tests/system/e2e.test.js

Hasil: 22/22 PASSED (100%)

Skenario	TC	Hasil
ST-01: Login & Navigasi	3	✓
ST-02: CRUD Siswa Lengkap	9	✓
ST-03: BK Case Flow	5	✓
ST-04: Audit Log	3	✓
ST-05: Dashboard Konsistensi	2	✓
TOTAL	22	✓ 100%

B.6 Hasil Black-Box Testing (EP & BVA)

File: eduscale-backend/tests/blackbox/blackbox.test.js

Hasil: 65/65 PASSED (100%)

Modul	Teknik	TC	Hasil
Login – Email	EP (4 kelas)	5	✓
Login – Password	EP (3 kelas) + BVA (5 nilai)	8	✓
Register – Nama	EP + BVA (6 nilai batas)	9	✓
Register – Role ID	EP + BVA (4 nilai batas)	9	✓
Paginasi – Page	BVA (3 nilai)	3	✓
Paginasi – Limit	BVA (4 nilai)	4	✓
Siswa – Nama/Status	EP	8	✓
Siswa – NIS	EP + BVA (6 nilai batas)	7	✓
Guru – NIP	BVA (3 nilai) + EP (2)	5	✓
BK Case – Status	EP (6 kelas)	6	✓
Duplikat Data	EP	1	✓
TOTAL		65	✓ 100%

B.7 Code Coverage

Jalankan `cd eduscale-backend && node_modules/.bin/jest --coverage` untuk menghasilkan coverage report.

Target: $\geq 70\%$ pada modul kritis

Modul Kritis: `src/middleware/, src/utils/, src/controllers/`

Modul	Perkiraan Coverage
<code>src/middleware/authMiddleware.js</code>	~95%
<code>src/middleware/roleMiddleware.js</code>	~90%
<code>src/utils/generateToken.js</code>	~100%
<code>src/utils/logActivity.js</code>	~80%
<code>src/controllers/authController.js</code>	~70%
<code>src/controllers/studentController.js</code>	~70%

B.8 Load Testing – Konfigurasi & Target

Tool: k6

Script: `tests/load/load_test.js`

Skenario	VU	Durasi	Threshold
Smoke Test	2	30 detik	Error rate < 5%
Load Test	50	2 menit	p95 response < 500ms
Stress Test	150	3 menit	p95 response < 1000ms
Spike Test	200	1 menit	Error rate < 10%

Cara Menjalankan:

```
brew install k6
cd eduscale-backend && npm start # Terminal 1
k6 run tests/load/load_test.js # Terminal 2
```

B.9 API Automation (Postman/Newman)

Collection: 03_Test_Scripts_and_Automation/
jmeter_or_postman_scripts/EduScale_API.postman_collection.json

Total Request: 28 request dengan assertions otomatis

Cara Menjalankan:

```
npm install -g newman newman-reporter-htmlextra
cd 03_Test_Scripts_and_Automation/jmeter_or_postman_scripts
./run_newman.sh
```

B.10 UI Automation (Playwright)

Folder: 03_Test_Scripts_and_Automation/ui_automation_playwright/

Total Skenario: 26 E2E skenario

Browser: Chromium, Firefox

Cara Menjalankan:

```
cd 03_Test_Scripts_and_Automation/ui_automation_playwright
npm install
npx playwright install chromium
npm test
```

B.11 Defect Summary

Total Defect	Critical	High	Medium	Low	Terbuka	Tertutup
1	0	0	1	0	0	1

DEF-001 (CLOSED): Test IT-028 timeout akibat `jwt.verify(undefined, secret)` pada Express 5.

Resolusi: Test case diubah untuk mengirim format JWT tidak valid yang terdefinisi dengan baik.

B.12 Kesimpulan

-  172/172 automated tests PASSED (100%)
-  Tidak ada defect terbuka

-  **Semua level pengujian tercakup:** Unit, Integration, System, Black-Box, UAT, Load, API Automation, UI E2E
 -  **Semua tools yang disyaratkan digunakan:** Jest, Supertest, Postman/Newman, k6, Playwright
 -  **Sistem siap untuk deployment production**
-

Disetujui oleh: Kelompok 7

Tanggal: 31 Juli 2026